

Sound Becomes a Signal

Project: Helpful Classroom Sorter

1. Sound Becomes a Signal

Teach the learning cycle

Primary target: Interpret an ordered sequence of sound-signal samples with a stated time interval

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Pictures Become Numbers

Without opening the old lesson, explain one key idea from Pictures Become Numbers. Give an example and a mistake that the idea helps you avoid.

Look for: Encode and reconstruct a small image using an explicit grid convention

Reference example, not the only acceptable wording: The upper row has black, white, black; the lower row has white, black, white. Reading back gives the supplied two rows. There are three black pixels among six total positions.

Sensors Are Messengers

Without opening the old lesson, explain one key idea from Sensors Are Messengers. Give an example and a mistake that the idea helps you avoid.

Look for: Describe a measurement using its quantity, value, unit and conditions

Reference example, not the only acceptable wording: Keep the observed value 24 and approximate time lunchtime; mark the unit, exact time if needed, location, exposure and device identity as unknown or to be checked. A suitable completed record needs air temperature, value, unit and relevant conditions. Invented degrees C or shade is not an observation.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A whisper recorded close to a microphone has larger sample amplitudes than a shout recorded far away. Does the recording prove the whisper was louder at its source? Propose one controlled follow-up.

Concept to explain

Computers receive changing measurements, not human-like hearing.

Term · Meaning

Term: waveform

Meaning: graph of a signal changing over time

Term: sample

Meaning: one measurement at one moment

Term: amplitude

Meaning: size or strength of a signal value

Term: noise

Meaning: unwanted signal mixed with the target

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Keep the distinction between physical pressure variation, microphone signal and stored numerical samples. The examples use arbitrary nonnegative signal values to avoid introducing signed waveforms before they are needed; they do not model real speech faithfully. The lesson establishes ordering, time metadata and information loss. It does not teach a sampling theorem or make hearing-health claims.

Readiness answer

The total does not preserve positions. Different arrangements can contain the same number of black pixels.

A time strip marked 0 through 8 milliseconds and paper sample-value cards

A cover sheet to hide omitted time positions

Printed fictional speech requests; recording students or opening accounts is unnecessary

End goal for the pictorial case

By the end, I can connect ordered sound samples to timing and explain what a maximum loses.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

INPUT changing air pressure

PROCESS sample the signal and compare a clap pattern

OUTPUT clap detected, not detected, or uncertain

CORE IDEA Computers receive changing measurements, not human-like hearing.

UNDER THE HOOD Audio is represented by samples taken over time. Sample rate describes how often measurements are taken; amplitude describes signal strength. Noise can overlap the target pattern, so detection requires a decision rule or model and an uncertainty policy.

Draw a fresh example with all three stages:

Every step connects a claim to observable evidence.

CASE A microphone converts air-pressure changes into a sequence of measured values that can be plotted as a waveform.

STEP 1 — OBSERVE changing air pressure

STEP 2 — TRACE sample the signal and compare a clap pattern

STEP 3 — CONCLUDE clap detected, not detected, or uncertain

MODEL REASONING Look for two strong peaks separated by the expected gap. Noise may add extra peaks or hide a clap. A safe detector needs thresholds, timing rules, and a not-sure result when evidence is weak.

DO NOT OVERCLAIM A microphone measures a signal; it does not understand a speaker's meaning by itself.

What new evidence could change this answer?

Concrete teaching case

A microphone records quiet and loud claps at the same distance and settings.

What waveform difference is expected? Does it reveal a spoken word?

Exact explanation to model

The louder clap generally produces larger amplitude. Amplitude is not word meaning. Distance, gain and noise must be controlled.

Explain the added worked example

Sound involves changing pressure in a medium such as air. A microphone responds to that variation and produces a changing signal. A digital recording stores measurements of the signal at successive times. The values are ordered: swapping their times changes the represented pattern.

We will use tiny invented sequences to study this idea on paper. Their values are arbitrary signal units, not decibels or a test of safe listening levels. Real audio uses many more samples, so our five-point examples

should not be played as if they were a realistic recording of speech.

A recording must include enough information to interpret the values, including their time spacing and encoding. The sequence 0,2,0,2,0 does not by itself say how quickly the changes happened. This is similar to needing image dimensions as well as pixel values.

There are five samples with four gaps between the first and last. Each gap is 1 millisecond, so the elapsed time from first to last is 4 milliseconds. If the same five values were spaced 2 milliseconds apart, they would span 8 milliseconds. Equal value lists with different timing do not describe the same time pattern.

Key terms

Term · Meaning

Term: Signal

Meaning: A changing physical or recorded quantity that can carry information

Term: Sample

Meaning: A recorded value of a signal at a particular time in this lesson

Term: Sampling interval

Meaning: The time between successive recorded samples

Term: Waveform

Meaning: A representation of how a signal value changes over time

Term: Transcript

Meaning: A text representation of spoken words, which may contain errors or omit sound details

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

The largest value in 0,2,0,2,0 is 2. The largest value in 0,0,2,0,0 is also 2. The first sequence reaches 2 twice at different times, while the second does so once. Keeping only the maximum loses that distinction, just as counting black pixels lost their arrangement.

Sparse measurements can also miss changes. If we inspect the first sequence only at times 0, 2 and 4 milliseconds, all retained values are 0. The two intervening values of 2 disappear from the record. It would be incorrect to conclude from these retained samples alone that the signal stayed at zero between them.

More frequent sampling can preserve faster changes under suitable recording conditions, but sampling frequency alone does not make every recording clear. Microphone placement, background sound, clipping and the recording format can also matter. We are studying one source of missing information, not a complete audio engineering specification.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Sound Becomes a Signal

A fictional microphone display records five samples during a short classroom tap.

The facts we will use

Times are 0,1,2,3,4 ms in order.

Signal A: 0,3,0,1,0. Signal B: 0,1,0,3,0. Toy values are signed measurement units; these happen to be nonnegative.

Same peak, different sample timing

	0 ms	1 ms	2 ms	3 ms	4 ms
Signal A	0	3	0	1	0
Signal B	0	1	0	3	0

Read a cell using both its row and column labels.

Sound Becomes a Signal

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Timed sampling

Associate every measurement with its sampling time.



2. Signal A

Its largest sample 3 occurs at 1 ms.



3. Signal B

Its largest sample 3 occurs at 3 ms.



4. Maximum summary

Both maxima are 3, but the strongest recorded tap arrives at different times.

CHANGE ONE THING

Retain only samples at 0,2,4 ms.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Timed sampling

What happens and why: Associate every measurement with its sampling time.

Step: 2. Signal A

What happens and why: Its largest sample 3 occurs at 1 ms.

Step: 3. Signal B

What happens and why: Its largest sample 3 occurs at 3 ms.

Step: 4. Maximum summary

What happens and why: Both maxima are 3, but the strongest recorded tap arrives at different times.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The sequences contain the same values in different orders. Keeping the maximum preserves peak recorded value while losing its timing. A graph should keep time on the horizontal axis; shuffling samples changes the recorded signal even if its maximum stays fixed.

What this example does not establish

These sparse toy values cannot establish perceived loudness, pitch or what happened between sampling times.

Change one thing in the pictured case

Change: Retain only samples at 0,2,4 ms.

Prediction: Both retained sequences are 0,0,0.

Reason: The selected times omit both nonzero samples from each signal.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: A microphone records quiet and loud claps at the same distance and settings.

Write the question to explain: What waveform difference is expected? Does it reveal a spoken word?

Reveal after predictions: The louder clap generally produces larger amplitude. Amplitude is not word meaning. Distance, gain and noise must be controlled.

Keep for an independent attempt: Two recordings contain the same word: one quiet and one loud. Which device receives the sound? Does a larger wave automatically mean a different word? Explain.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

The interval is 2 ms and first-to-last duration 8 ms. Keeping 0,4,8 ms retains 0,0,0 and misses the 3 and 1 at 2 and 6 ms. A transcript can preserve estimated words but may omit pauses or background sounds. Correct arithmetic on a transcribed number does not establish that the number matches the recording.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start 0 to 7 minutes

Teacher action: Retrieve equal pixel counts with different arrangements. Replace position in space with position in time.

Stage: 7 to 20 minutes

Teacher action: Trace the recording/transcript/processing diagram. Explain the invented units and times.

Stage: 20 to 33 minutes

Teacher action: Count gaps physically between five sample cards. Compare the same values at two time intervals.

Stage: 33 to 45 minutes

Teacher action: Complete W1 and W2, covering omitted positions and naming lost changes.

Stage: Session 2 start 0 to 8 minutes

Teacher action: Compare the equal-maximum sequences. Require a specific difference, not only less information.

Stage: 8 to 23 minutes

Teacher action: Complete W3 and practise constructing two records that agree at retained times.

Stage: 23 to 38 minutes

Teacher action: Collect W4 independently, checking all seven positions and the three retained constraints.

Stage: 38 to 45 minutes

Teacher action: Discuss W5 and the exit question. Close the unit by connecting evidence, representation, measurement, fitting, pixels and samples.

Reduce help deliberately

Completion task: Retain only times 0,4,8 from W1. Write the retained values. Give one claim about the original signal that these retained values do not justify.

Teacher check: The retained values are 0,0,0. They do not justify saying the signal was always zero or had no changes between those times; the original had values 3 and 1 at omitted times.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Sound as sampled information. Proposed explanation: “Five samples make five gaps”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Place five cards in a row and point to each space between them. The span ends at the last sample.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Does the all-zero retained record prove the full signal was silent? Use the picture.

Reasoned teacher answer

No. The full supplied signals contain nonzero values at 1 and 3 ms; those measurements were discarded.

Changed-case reasoning: Both retained sequences are 0,0,0. The selected times omit both nonzero samples from each signal.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Predict first, test honestly, and explain what the evidence means.

State your prediction before the result is known.

Complete the task: Build a clap-pattern detector; test it with background noise.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

CLAIM TO REPAIR Louder always means a different word.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Five samples make five gaps: Place five cards in a row and point to each space between them. The span ends at the last sample.

Equal recorded zeros prove the signal stayed zero: Reveal hidden sample cards between retained zeros. Construct both changing and unchanging originals.

A transcript is the recording in another file type: Compare the information each representation retains. Text estimates words and need not preserve sound details or exact wording.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed

incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

The louder clap generally produces larger amplitude. Amplitude is not word meaning. Distance, gain and noise must be controlled.

Independent question

Two recordings contain the same word: one quiet and one loud. Which device receives the sound? Does a larger wave automatically mean a different word? Explain.

Expected independent answer

A microphone receives the sound. Larger amplitude can mean louder sound; it does not by itself establish a different word. Word recognition needs the sound pattern over time.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “Louder always means a different word.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

No. Recorded amplitude depends on distance, gain and background as well as source sound. Compare sounds at the same distance and gain in similar conditions. Even then amplitude does not identify the words or their meaning.

Explanations for the additional workbook examples

W1 Read samples and gaps

Five samples have values 0,3,0,1,0 at times 0,2,4,6,8 milliseconds. State the sample count, interval and elapsed time from first to last. Explain why there are four gaps.

Five samples, a 2 ms interval, and 8 ms from first to last. Consecutive pairs make gaps 0-2, 2-4, 4-6 and 6-8: four gaps between five samples.

W2 Inspect what was discarded

Retain only times 0,4,8 from W1. Write the retained values. Give one claim about the original signal that these retained values do not justify.

The retained values are 0,0,0. They do not justify saying the signal was always zero or had no changes between those times; the original had values 3 and 1 at omitted times.

W3 Compare equal summaries

Sequences A=0,2,0,2,0 and B=0,0,2,0,0 use the same time interval. State their maximum values and one difference the maximum discards. Name information that would preserve that difference.

Both maxima are 2. A reaches 2 at two sample positions while B does so at one; their timing patterns differ. The full ordered samples with timing preserve the distinction. A maximum alone does not.

W4 Construct a new hidden change

A record keeps values 0 at times 0,3,6 milliseconds. Construct two seven-sample sequences for every whole-millisecond time from 0 through 6 that agree with those retained values but differ at omitted times. Explain why the retained record cannot prove that nothing changed between measurements.

One pair is [0,0,0,0,0,0,0] and [0,2,1,0,3,1,0]. Both have zero at indices/times 0,3,6, but the second changes at omitted times. The retained record is compatible with both, so it cannot establish an unchanged intervening signal. Any valid seven-value pair with those constraints is acceptable.

W5 Check a speech pipeline

A recording says Set the timer for fourteen minutes. A transcript says forty minutes and the timer is set to 40. Name the first mismatch, propose a confirmation step and explain why the transcript cannot be treated as a complete copy of the recording.

The first stated mismatch is fourteen becoming forty in transcription. Show or read back the interpreted duration and require correction/confirmation before starting the timer. A transcript may misstate words and omit tone, pauses and background sounds; correct handling of 40 does not validate the earlier transcription.

Assess independent transfer

Selected case: A record keeps values 0 at times 0,3,6 milliseconds. Construct two seven-sample sequences for every whole-millisecond time from 0 through 6 that agree with those retained values but differ at omitted times. Explain why the retained record cannot prove that nothing changed between measurements.

Selected case answer

One pair is [0,0,0,0,0,0,0] and [0,2,1,0,3,1,0]. Both have zero at indices/times 0,3,6, but the second changes at omitted times. The retained record is compatible with both, so it cannot establish an unchanged intervening signal. Any valid seven-value pair with those constraints is acceptable.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Helpful Classroom Sorter

If the sorter uses sound, specify the exact sound pattern needed and test whether classroom noise makes that input unreliable.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two mechanisms with the same visible output and explain what evidence distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 7: Categories and Labels. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Use large labelled boxes and tactile counters. Read values aloud while pointing to their time labels. No hearing-based task is required. Extension: ask whether doubling the interval changes the number of samples, the first-to-last duration or the stored values when the same list is replayed under the new timing convention.

W4 is secure when each proposed sequence has seven values, both satisfy the zero constraints at 0,3,6, at least one omitted position differs and the explanation identifies ambiguity between measurements. If positions are confused, reteach with labelled time boxes, then assess a fresh retained-time pattern.

Week 7 builds on this unit by defining categories operationally so different people can label the same evidence consistently.

Plan the delayed return

Suggested return points in this level: ai-1/7, ai-1/9, ai-1/12. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Record the sorter's sampling interval and retain relevant sequences when a decision depends on changes over time.

7. Prepare, teach and assess

Teaching target: Interpret an ordered sequence of sound-signal samples with a stated time interval

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Retain only times 0,4,8 from W1. Write the retained values. Give one claim about the original signal that these retained values do not justify.

Reasoned practice answer: The retained values are 0,0,0. They do not justify saying the signal was always zero or had no changes between those times; the original had values 3 and 1 at omitted times.

Check every learner before discussion: Use the worked case for Sound as sampled information. Proposed explanation: “Five samples make five gaps”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Place five cards in a row and point to each space between them. The span ends at the last sample.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Retain only samples at 0,2,4 ms. Require a prediction and a reason before comparing.

Expected result: Both retained sequences are 0,0,0. Reason: The selected times omit both nonzero samples from each signal.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.